Report

MEDOPAD

Empowering Health, Embracing Tomorrow.

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Course: Machine Learning and Deep Learning

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1. Introduction

Medopad is an application that uses artificial intelligence to detect 7 different diseases, offering users a practical solution for regular monitoring of their health. This application has been designed to improve the accessibility and convenience of medical testing for users, while improving the efficiency of the healthcare system through early and accurate disease detection

2. Objectives of the Application

Objectives: The primary objectives of MedoPad can be outlined as follows:

1. **Disease Detection:**
    - **Objective:** Enable accurate and reliable detection of seven major diseases COVID-19, Brain Tumors, Breast Cancer, Alzheimer's Disease, Diabetes, Pneumonia, and Heart Diseases—utilizing a combination of custom-made Convolutional Neural Networks (CNN), VGG-16, Random Forest, and XGBoost models.
2. **Accessibility:**
    - **Objective:** Democratize healthcare by providing individuals, irrespective of their geographic location or proximity to medical facilities, with immediate and remote access to disease detection services. This aligns with the broader goal of enhancing healthcare accessibility for diverse populations.
3. **User-Friendly Interface:**
    - **Objective:** Design an intuitive and user-friendly interface to facilitate seamless interaction. The interface should cater to individuals with varying levels of technological proficiency, ensuring ease of understanding and navigation.
4. **Future Enhancements:**
    - **Objective:** Continuously evolve and enhance the application by incorporating future improvements. This may include data enrichment to enhance model accuracy, the introduction of new disease detection capabilities based on X-ray scans or numerical inputs, and the integration of features providing personalized health guidance.
5. **Record-Keeping:**
    - **Objective:** Implement a comprehensive record-keeping system within the application. This ensures accurate and secure maintenance of disease detection results, contributing to a holistic approach to personalized healthcare.

In summary, MedoPad operates in the context of advancing healthcare through the integration of AI-driven diagnostics. Its objectives center around accurate disease detection, enhanced accessibility, user-friendly interactions, continuous improvement, robust record-keeping, and user education. These aspects collectively contribute to the application's mission of revolutionizing healthcare by merging the power of AI with compassionate and personalized medical solutions.

3. Data

In the context of MedoPad, each disease targeted by the application is associated with its own set of specific data parameters. This tailored approach ensures that the disease detection models, ranging from COVID-19 and Brain Tumors to Alzheimer's Disease and Diabetes, are finely tuned to the characteristics and intricacies of each medical condition. These parameters may include unique features identified by custom-made Convolutional Neural Networks (CNN), VGG-16, Random Forest, and XGBoost models. The individualized data for each disease contributes to the accuracy and reliability of the detection process, allowing MedoPad to provide specialized insights and information tailored to the nuances of different health conditions.

3.1 Covid-19

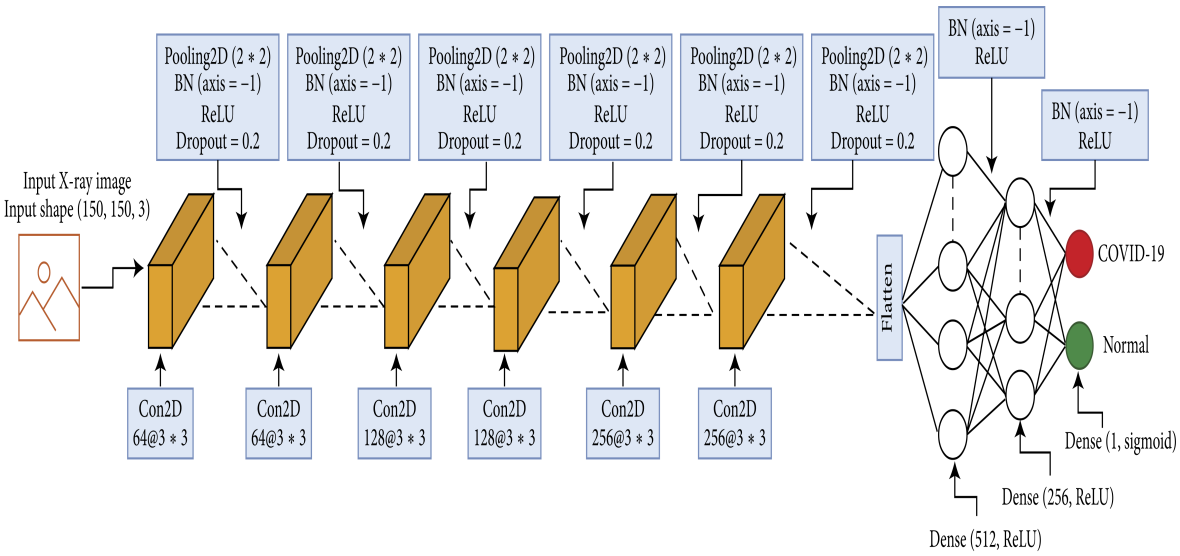
COVID data are collected from different publicly accessible datasets, online sources, and published papers.

Formats: All the images are in Portable Network Graphics (PNG) file format, and resolutions are 299x299 pixels. Click Here To See Dataset.

Normal Images: 10,192 Normal data are collected from three different datasets.

- 2473 CXR images are collected from padchest dataset.
- 183 CXR images from a Germany medical school.
- 559 CXR images from SIRM, Github, Kaggle, and Twitter.
- 400 CXR images from another Github source.
- 8851 RSNA.
- 1341 Kaggle.

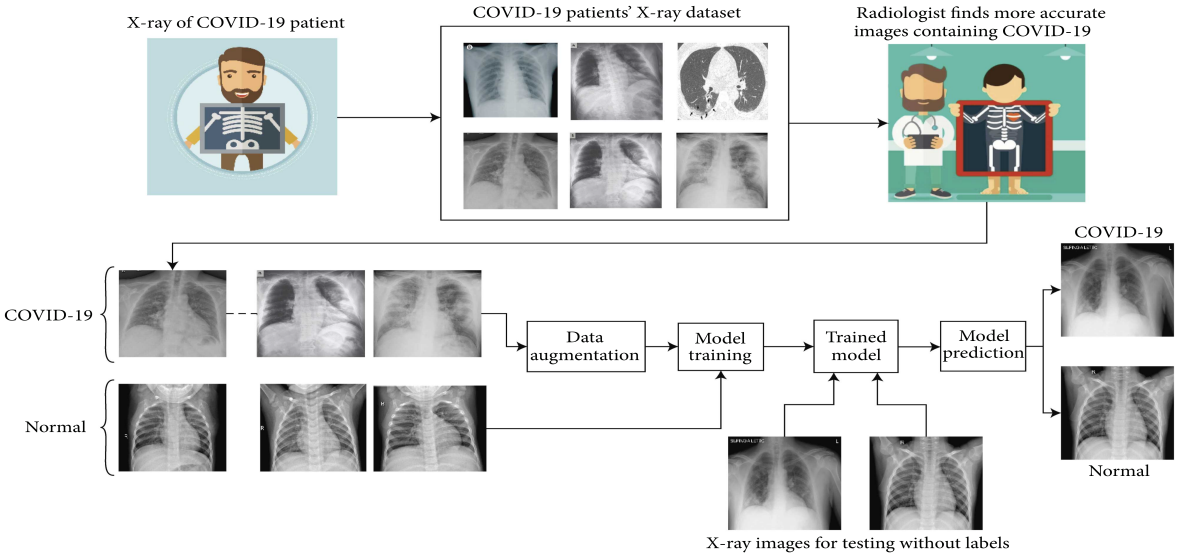
Model Architecture: CNN model consists of three building blocks as shown in This Fig:



Convolution Layer: These layers are responsible for learning the high-level features of an input image. There can be multiple convolution layers that can learn the image features in an incremental way.

Pooling Layer: It reduces the spatial size of the convolved volume, thus decreasing the computational power needed to process the data. It is also useful to extract the dominant features. This layer also performs noise suppression.

Fully Connected or Dense Layer: Once the size of the input volume is efficiently reduced by pooling layers, fully connected layers are used for learning complex and nonlinear combinations of high-level features. Fully connected layers typically work as an Artificial Neural Network (ANN) model, which is fed a one-dimensional vector formed from the output of the pooling layer.



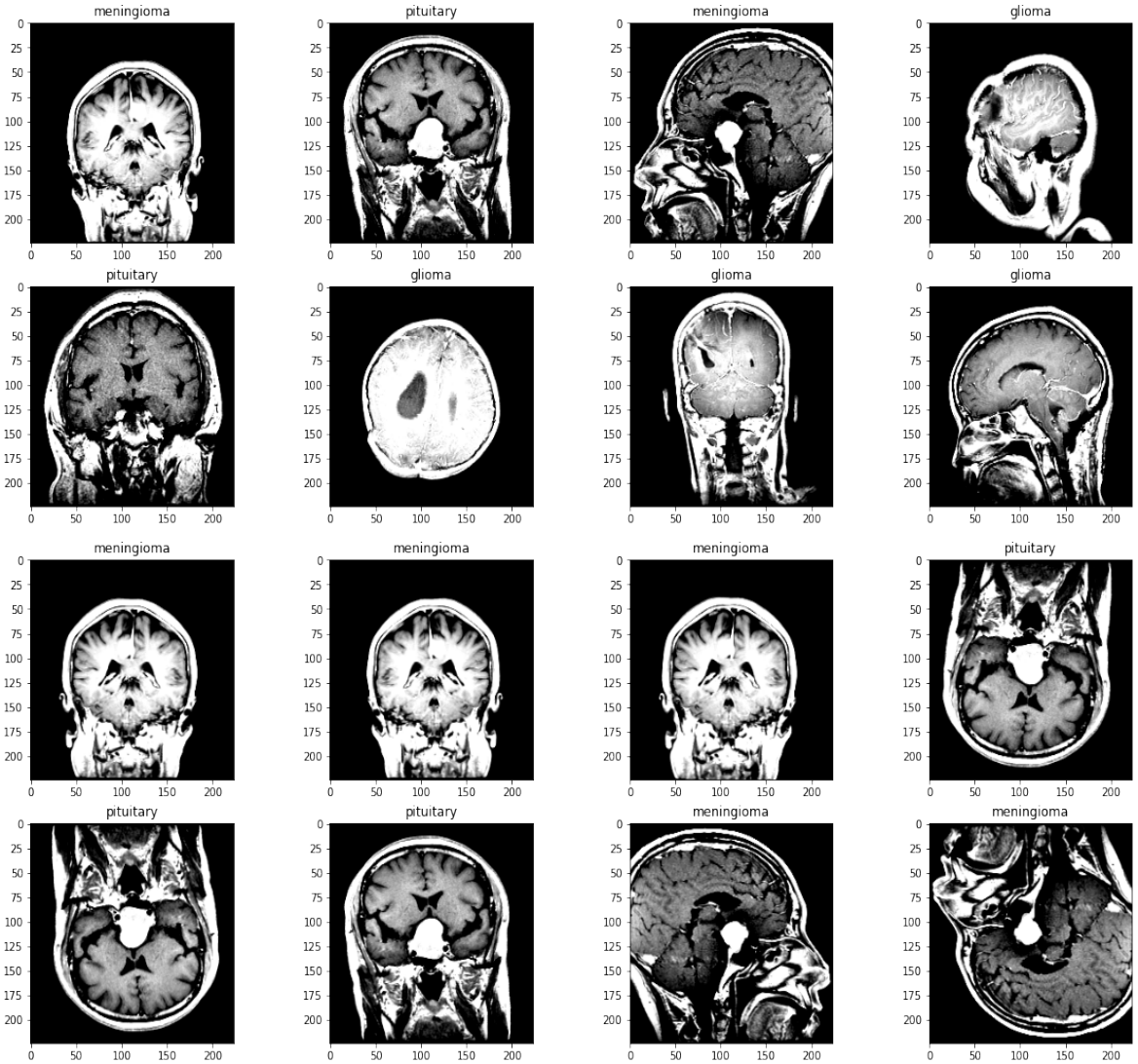
Result: The accuracy achieved was around 93 Percent.

3.2 Brain tumor

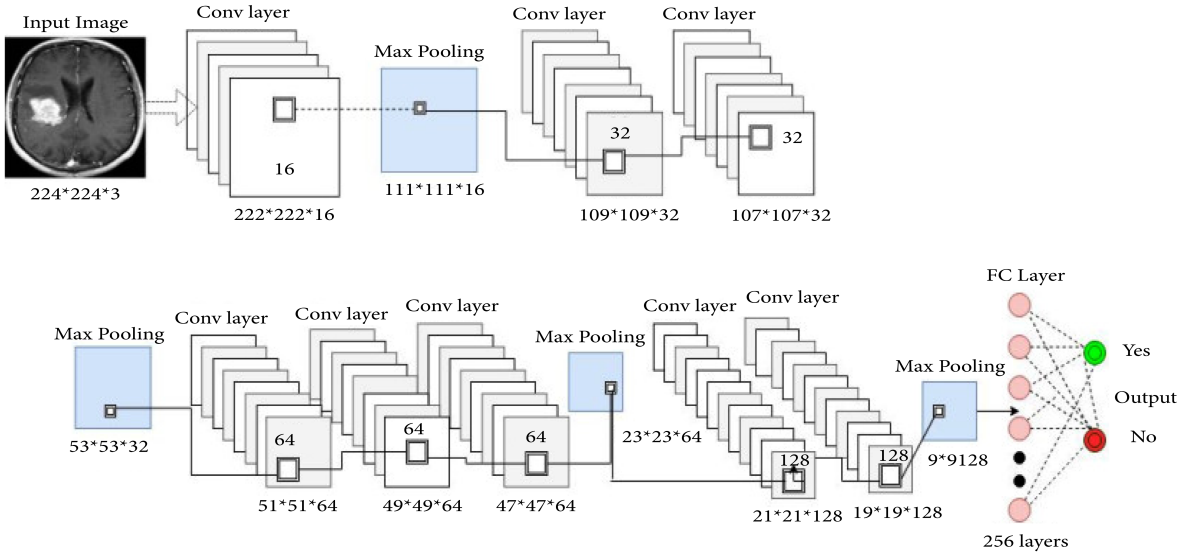
Brain Tumor data are collected from different publicly accessible datasets, online sources, and published papers.

Formats: All the images are in MRI FormClick Here To See Dataset.

Normal Images: There Are Data Before Augmentation and Data After Augmentation And here is the diffirence Between Them



Model Architecture: CNN model consists of three building blocks as shown in This Fig:

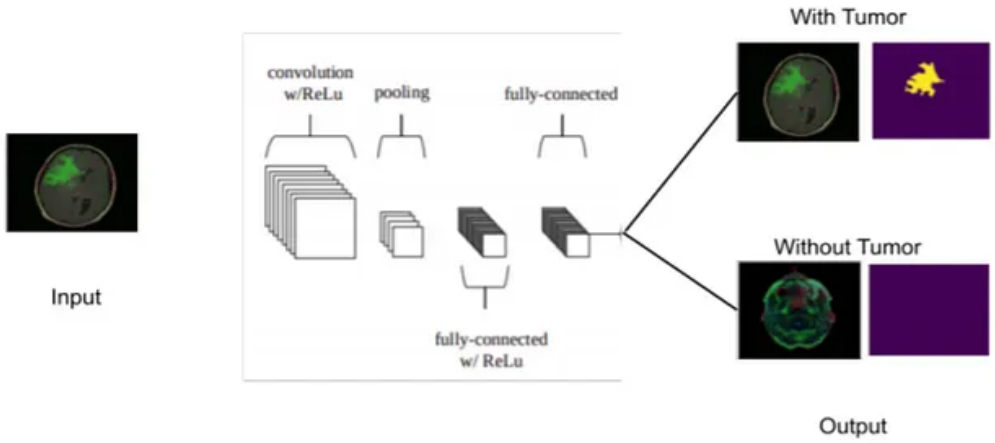


The input layer will hold pixel values of the MRI scan of brain. **The Convolution layer** will determine the weather the brain consist of tumor or not. These convolution layers are connected to local regions of the input through the calculation of the scalar product between their weights and the region connected to the input volume. The rectified linear unit (ReLu) focuses to apply an activation function such as sigmoid to the output of the activation produced by the previous layer.

The pooling layer will perform the same task down sampling along the spatial dimensionality of the given input; reducing the parameter number within that activation.

The fully connected layers will perform the task to produce class score from the activation, to be used for classification.

We Can Also work with ResNet50 Instead of building a new network, the Resnet50 model is used as the base to train the model. The residual network is a deep convolution neural network model that was introduced by Microsoft in 2015



Result: The accuracy achieved was around 99 Percent.

3.3 Breast cancer

Features are computed from a digitized image of a fine needle aspirate (FNA) of a breast mass. They describe characteristics of the cell nuclei present in the image.

Features Informations: Click Here To See Dataset.

- ID number.
- Diagnosis (M = malignant, B = benign) 3-32).
- radius (mean of distances from center to points on the perimeter).
- texture (standard deviation of gray-scale values).
- perimeter.
- area.
- smoothness (local variation in radius lengths).
- compactness (perimeter*perimeter / area - 1.0).
- concavity (severity of concave portions of the contour).
- concavity (severity of concave portions of the contour).
- symmetry.
- fractal dimension ("coastline approximation" - 1).

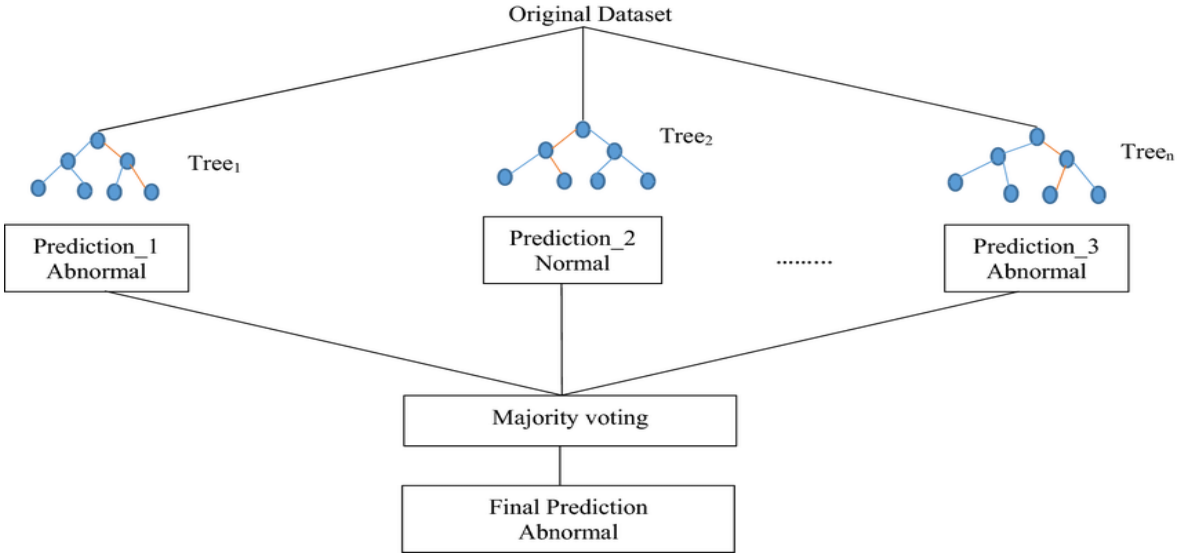
Explication: The mean, standard error and "worst" or largest (mean of the three largest values) of these features were computed for each image, resulting in 30 features. For instance, field 3 is Mean Radius, field 13 is Radius SE, field 23 is Worst Radius. All feature values are recoded with four significant digits.Missing attribute values: none . Class distribution: 357 benign, 212 malignant

Model Architecture: Random Forest model consists of Two steps:

Step One: You draw B samples with replacement from the original data set where B is a number less than or equal to n, the total number of samples in the training set.

Step Two: Train a decision trees on newly created bootstrapped samples. Repeat the Step1 and Step2 any number of times that you like. Generally, higher the number of trees, the better the model. But remember! Excess number of trees can make a model complicated and ultimately lead to overfitting as your model starts seeing relationships in the data that do not exist in the first place.

Explication Random forest is different from the vanilla bagging in just one way. It uses a modified tree learning algorithm that inspects, at each split in the learning process, a random subset of the features. We do so to avoid the correlation between the trees.



Result: The accuracy achieved was around 91 Percent.

3.4 The Alzheimer's

The Data is collected from several websites/hospitals/public repositories.

Formats: All the images are resized into 128 x 128 pixelsClick Here To See Dataset.

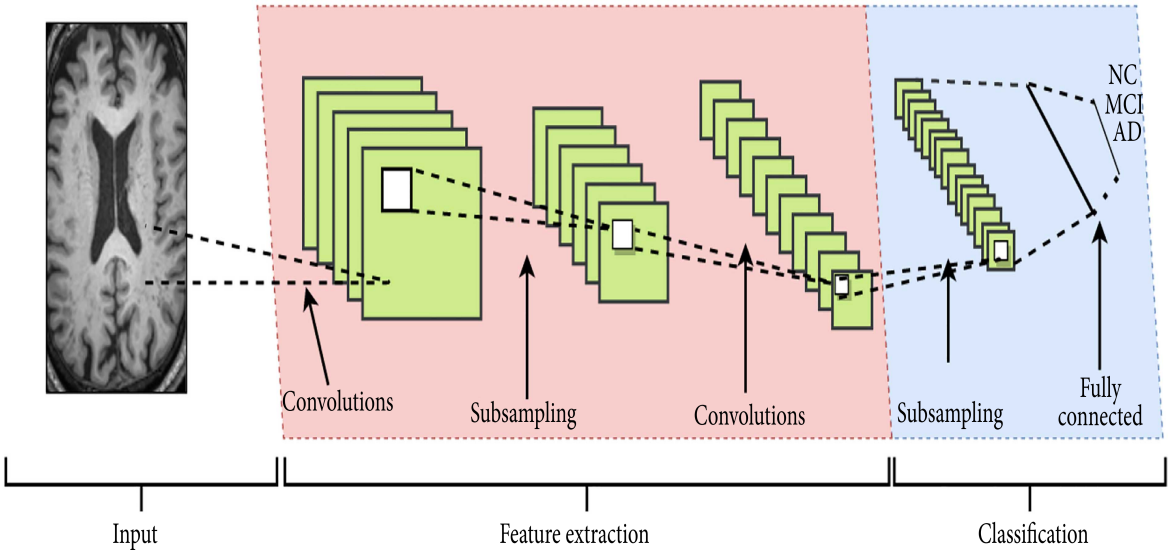
About Dataset:

- The Dataset is consists of total 6400 MRI images.
- The Dataset has four classes of images.
- All the images are resized into 128 x 128 pixels.
- The Dataset is consists of Preprocessed MRI (Magnetic Resonance Imaging) Images.

The Dataset has four classes of images:

- Class - 1: Mild Demented (896 images)
- Class - 2: Moderate Demented (64 images)
- Class - 3: Non Demented (3200 images)
- Class - 4: Very Mild Demented (2240 images)

Model Architecture: CNN model consists of three building blocks as shown in This Fig:



The input layer will hold pixel values of the MRI scan of brain. **The Convolution layer** will determine the weather the brain consist of tumor or not. These convolution layers are connected to local regions of the input through the calculation of the scalar product between their weights and the region connected to the input volume. The rectified linear unit (ReLu) focuses to apply an activation function such as sigmoid to the output of the activation produced by the previous layer.

The pooling layer will perform the same task down sampling along the spatial dimensionality of the given input; reducing the parameter number within that activation.

The fully connected layers will perform the task to produce class score from the activation, to be used for classification.

Result: The accuracy achieved was around 73 Percent.

3.5 Diabetes

This dataset is originally from the National Institute of Diabetes and Digestive and Kidney Diseases. The objective is to predict based on diagnostic measurements whether a patient has diabetes.

Features Informations: Click Here To See Dataset.

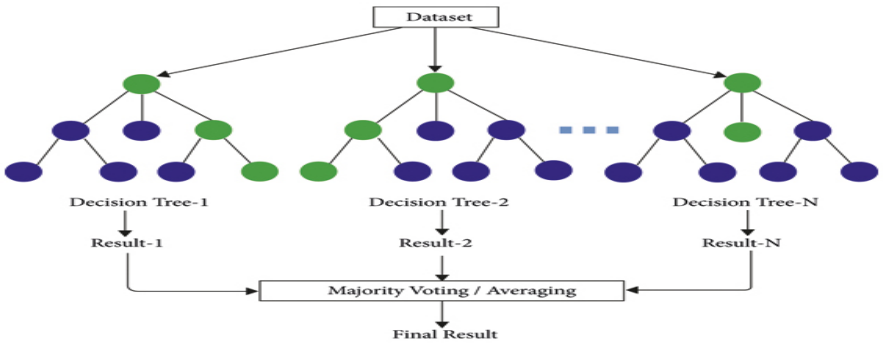
- Pregnancies: Number of times pregnant.
- DGlucose: Plasma glucose concentration a 2 hours in an oral glucose tolerance test.
- BloodPressure: Diastolic blood pressure (mm Hg).
- SkinThickness: Triceps skin fold thickness (mm).
- Insulin: 2-Hour serum insulin (mu U/ml).
- BMI: Body mass index (weight in kg/(height in m)*m).
- DiabetesPedigreeFunction: Diabetes pedigree function.
- Age: Age (years).
- Outcome: Class variable (0 or 1).

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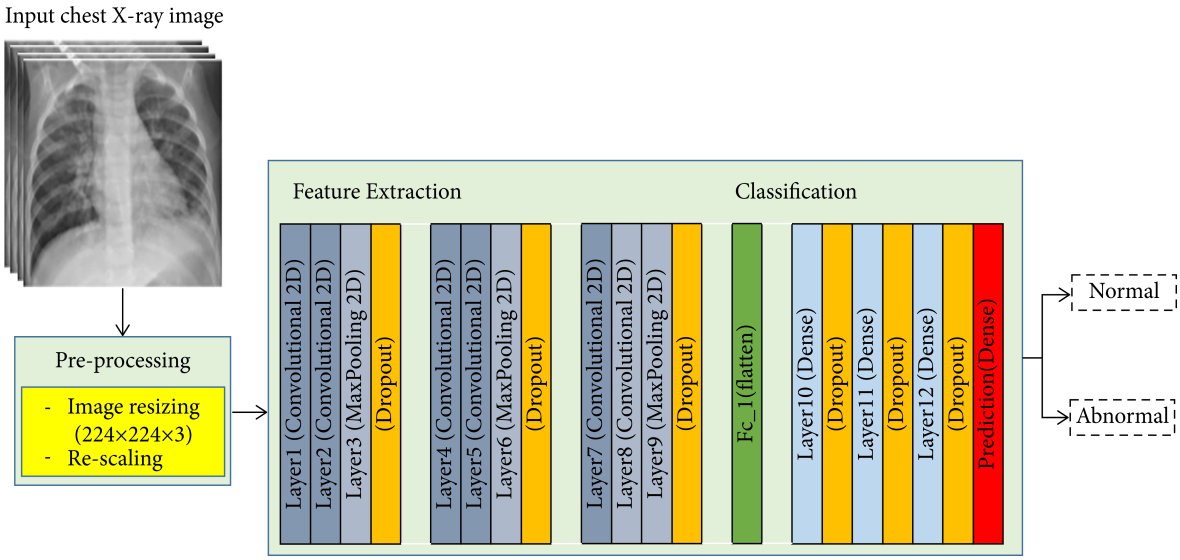
Result: The accuracy achieved was around 66 Percent.

3.6 Pneumonia

Pneumonia data are collected from different publicly accessible datasets, online sources, and published papers.

Formats: he dataset is organized into 3 folders (train, test, val) and contains subfolders for each image category (Pneumonia/Normal). There are 5,863 X-Ray images (JPEG) and 2 categories (Pneumonia/Normal). Chest X-ray images (anterior-posterior) were selected from retrospective cohorts of pediatric patients of one to five years old from Guangzhou Women and Children's Medical Center, Guangzhou. All chest X-ray imaging was performed as part of patients' routine clinical care. For the analysis of chest x-ray images, all chest radiographs were initially screened for quality control by removing all low quality or unreadable scans. The diagnoses for the images were then graded by two expert physicians before being cleared for training the AI system. In order to account for any grading errors, the evaluation set was also checked by a third expert. Click Here To See Dataset.

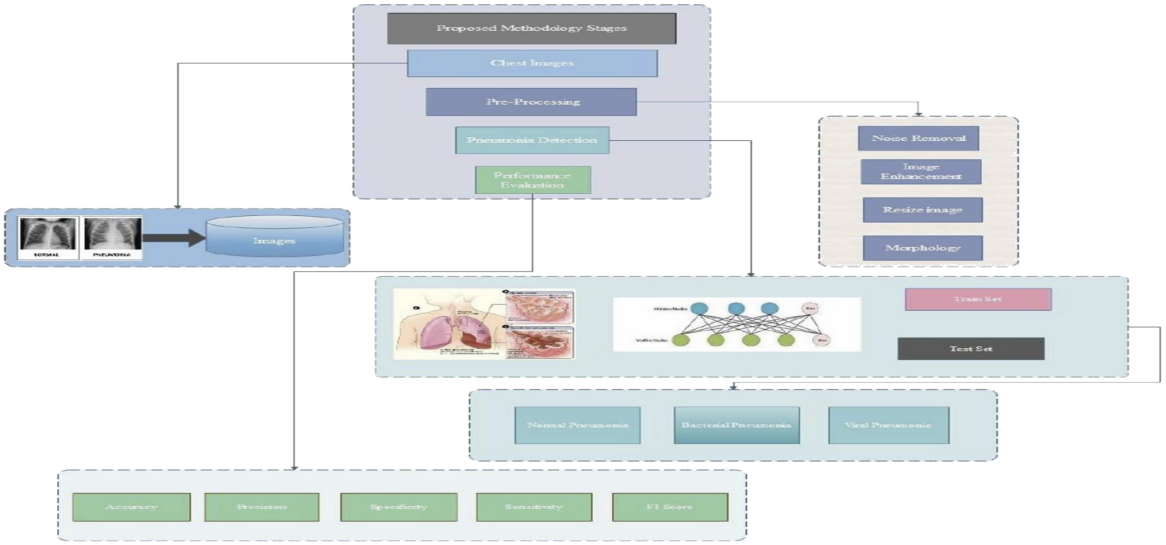
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Result: The accuracy achieved was around 92 Percent.

3.7 Heart Disease

This data set dates from 1988 and consists of four databases: Cleveland, Hungary, Switzerland, and Long Beach V. It contains 76 attributes, including the predicted attribute, but all published experiments refer to using a subset of 14 of them. The "target" field refers to the presence of heart disease in the patient. It is integer valued 0 = no disease and 1 = disease.

Features Informations: Click Here To See Dataset.

- age.
- sex.
- chest pain type (4 values).
- resting blood pressure.
- serum cholestoral in mg/dl.
- fasting blood sugar ¿ 120 mg/dl.
- resting electrocardiographic results (values 0,1,2).
- maximum heart rate achieved.
- exercise induced angina.
- oldpeak = ST depression induced by exercise relative to rest.
- the slope of the peak exercise ST segment.
- number of major vessels (0-3) colored by flourosopy.
- thal: 0 = normal; 1 = fixed defect; 2 = reversable defect The names and social security numbers of the patients were recently removed from the database, replaced with dummy values.

Model Architecture: XGBoost is a high-performance version of the Gradient Boosting algorithm optimized with various modifications.It was started to be used by Tianqi Chen and Carlos Guestrin in the article "XGBoost: A Scalable Tree Boosting System" in 2016. XGBoost has a similar working logic to Gradient Boosting, but there are some differences and advanteges

Steps:

The first step in XGBoost is to make the first guess, that is, to determine the base score. The base score is usually set at 0.5. So the initial prediction values are 0.5. Then the residuals are obtained by subtracting 0.5 from the actual y values. A tree model is established with these residues obtained as in GBM..

Similarity Score = (Sum of Residuals)*(Sum of Residuals) / (number of Residuals +)

Gain = Left Similarity + Right Similarity - Root Similarity

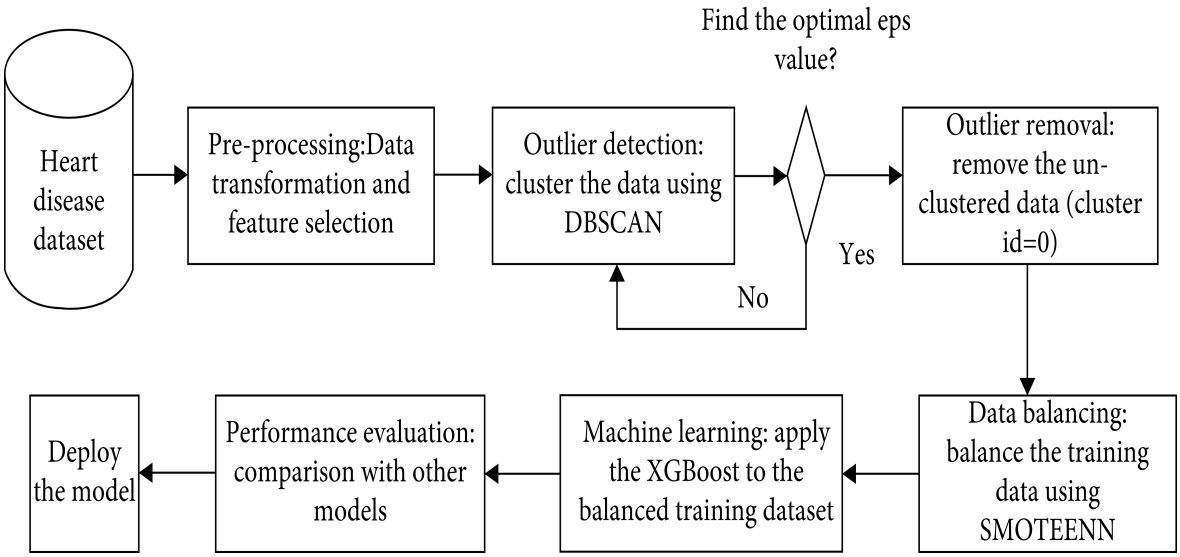
Since calculating these similarity and gain scores would take a long time on large datasets, xgboost divides the data into quantiles instead of examining each value in the data.

The default number of quantile is 33. As the number of quantiles increases, xgboost will look at smaller ranges and make better predictions, but at the same time the training time will be longer.An algorithm called "Sketches" is used to overcome this training time problem. The "Sketches" algorithm converges to find the quantiles.

After the tree with the highest gain score is determined, the pruning process is started.During the pruning process, the "gamma" hyperparameter is used as a metric. if Gain Score ¡ Gamma,the branch is pruned.As gamma increases, the most valuable branches remain on the tree and This pruning helps prevent overfitting.Pruning is done from the bottom to the top. If the bottom branch is not pruned, the upper branches are not examined.

Prediction is made after the pruning process is completed.The tree prediction is multiplied by the learning rate and added to the prediction value of the first tree, and a new prediction value is formed. These operations continue until the specified number of iterations, namely n estimators (number of boosting trees).

New Prediction = First Prediction + (Learnin Rate)* (Second Prediction)



Result: The accuracy achieved was around 98 Percent.

4. Future Scope

- As time passes, i will be available with more and more data and i will try to make these models even more accurate by training on much more data. -Also, i will be adding more disease detections that can be detected using X-ray scans or just by inputting simple numbers. -I am also planning to add more features like if a person is found positive then the app will show him what precautions he needs to take and how he can cure himself. -I will also be storing the detection records. -So these are some future improvements/additions i plan to add.

5. Conclusion

MedoPad isn't just a project; it's a vision for the future of healthcare, where AI empowers individuals to access their health status conveniently and promptly from the comfort of their homes. Together, we're taking a significant step towards this exciting future of AI-driven healthcare."

6. Scan This QR For Github Repo



Bibliography